

COMP6203 - INTELLIGENT AGENTS

Introduction to Intelligent Agents

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Intelligent Agents

OUTLINE

- Topics covered today
 - Towards multi-agent systems: trends in Computer Science
 - What is an agent?
 - What makes an agent intelligent?
 - Multi-agent systems
 - Agents and related concepts

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OVERVIEW

- Five ongoing trends have marked the history of computing:
 - *Ubiquity*
 - *Interconnection*
 - *Intelligence*
 - *Delegation*
 - *Human-orientation*

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UBIQUITY

- The continual reduction in cost of computing capability has made it possible to introduce processing power into places and devices that would have once been uneconomic
- As processing capability spreads, sophistication (and intelligence of a sort) becomes ubiquitous
- Increased amount of data becoming available, and processing of this data is increasingly done at the edge of the network, so-called edge computing

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INTERCONNECTION

- Computer systems today no longer stand alone, but are networked into large distributed systems
- Distributed and concurrent systems have become the norm: researchers are putting forward theoretical models that portray computing as primarily a process of interaction

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INTELLIGENCE

- The complexity of tasks that we are capable of automating and delegating to computers has grown steadily
- Many of these tasks are ones that can be thought of as requiring a good deal of intelligence.

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DELEGATION

- Computers are doing more for us – without our intervention
- We are giving control to computers, even in safety critical tasks, e.g.:
 - *Fly-by-wire aircraft*, where the machine's judgment may be trusted more than an experienced pilot
 - *Autonomous cars and public transport*
 - *AI Agents: scheduling, planning, task automation, research, etc.*

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HUMAN ORIENTATION

- AI programs need input from humans: e.g. understanding preferences and for controlling and checking decisions
- Explainable AI: Need higher level of abstractions to interact with complex systems, using familiar metaphors: cannot expect to understand how 'engine' works to use it
- AI is not only about automation, but about augmenting human's intelligence

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WHERE DOES THIS BRING US?

- Delegation and Intelligence imply the need to build computer systems that can act effectively on our behalf
- This implies:
 - The ability of computer systems to act independently
 - The ability of computer systems to act in a way that represents our best interests while interacting with other humans or systems
 - In turn, this implies systems that can cooperate and reach agreements (or even compete) with other systems that have different interests (much as we do with other people)
- Together, these trends have led to the emergence of a new field in computer science: **multi-agent systems**

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WHAT IS AN AGENT?

- *“A computer system capable of autonomous action in some environment, in order to achieve its delegated goals”*
- The key points are:
 - *Autonomy*: capable of independent action without need for constant intervention
 - *Delegation*: acts on behalf of its user or owner
- We think of an agent as being in a close-coupled, continual interaction with its environment: sense — decide — act — sense — decide . . .

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A SIMPLE AGENT (EXAMPLE)

- (Traditional) Thermostat
- Delegated goal is to maintain room temperature
- Perception is the heat sensor
- Actions are switch heat on/off
- Trivial decision-making



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INTELLIGENT BEHAVIOUR

- We typically think of *intelligent agents* as those exhibiting at least the following 3 types of behaviour:
 - *Reactive*
 - *Pro-active*
 - *Social*

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REACTIVE COMPUTER SYSTEMS

- If a program's environment is guaranteed to be fixed, a program can just execute blindly
- The real world is not like that: most environments are dynamic
- Software is hard to build for dynamic domains: program must take into account possibility of failure — ask itself whether it is worth acting!
- A reactive system is one that maintains an ongoing interaction with its environment, and responds to changes that occur in it (in time for the response to be useful); i.e. in real time

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PROACTIVE COMPUTER SYSTEMS

- Reacting to an environment can be easy; e.g. stimulus -> response rules
- But we generally want agents to do things for us
- Hence *goal-directed behaviour*
- Pro-activeness = generating and attempting to achieve goals; not driven solely by events; taking the initiative
- Recognising opportunities

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SOCIAL COMPUTER SYSTEMS

- The real world is a multi-agent environment: we cannot go around attempting to achieve goals without taking others into account
- Some goals can only be achieved by interacting with others
- Social ability in agents is the ability to interact with other agents and (usually) humans via cooperation, coordination, and negotiation
- At the very least, it means the ability to communicate..

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OTHER PROPERTIES

- Rationality: agents will act in order to maximise (expected) utility
- Learning/adaptation: agents improve performance over time

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CHALLENGES

- How to represent goals and user preferences
- How to optimise the decision making
- How to learn and adapt and improve over time

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MULTI-AGENT SYSTEMS

- A multi-agent system is one that consists of a number of agents interacting with each other
- In the most general case, agents will be acting on behalf of users with different goals and motivations with one-another
- To successfully interact, they will require the ability to cooperate, coordinate, and negotiate with each other, as much as people do

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MAS DESIGN CHALLENGES

- How can cooperation emerge in societies of self-interested agents?
- How can self-interested agents recognise conflict, and how can they (nevertheless) reach agreement?
- How can autonomous agents coordinate their activities so as to cooperatively achieve goals?

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COLLABORATIVE VS SELF-INTERESTED MAS

- ***Collaborative***: E.g. agents owned by the same person/organisation
 - Agent design can be controlled and made to coordinate
 - More robust compared to “centralised” system
 - Agents easy to control: No incentive needed
- ***Self-Interested/Competitive***: Each agent works on behalf of someone else
 - Each agent design is different; no control of design of other agents
 - Each agent pursues own goals
 - Still need to cooperate
 - Need mechanisms for resolving conflicts of interest

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LINKS WITH OTHER DISCIPLINES

- The field of intelligent agents and multi-agent systems is highly interdisciplinary
 - Robotics
 - Complex Systems Science
 - Economics
 - Philosophy
 - Game Theory
 - Logic
 - Social Sciences
 - AI
- This can be both a strength (infusing well-founded methodologies into the field) and a weakness (there are many different views as to what the field is about)

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SUMMARY

- This has been a brief introduction to Intelligent Agents and Multi-Agent Systems.
- We have argued that MAS are a natural development of computer science
- Next week: Utility and Strategic-form Games